StudySapiens Complete Chapter Notes

Chapter 5 — Enzymes

Complete Chapter Notes · Textbook topics converted into board-style long questions, with important conceptual short questions inside every topic · PECTAA 2025–26, Punjab English Medium

This chapter follows the textbook topics 5.1 to 5.6. Each topic is converted into board-style long questions, and each topic ends with conceptual short questions. Direct definitions are omitted.

Board pattern (Punjab 2026, new syllabus): Chapter 5 carries 1 MCQ and 3 short questions in Q.3. Shorts are 2 marks; longs are 8 marks.

TOPIC 5.1 — ENZYMES

Textbook topic 5.1 · printed pages 127–128

Q.1 Explain the nature of enzymes and the importance of the active site.

Board long question · 8 marks

Enzymes are specific proteins that speed up specific chemical reactions by lowering the required **activation energy**, but are unaltered themselves in the process. They are known as **biocatalysts**. Rates of enzyme-catalysed reactions may be 10^6 to 10^{12} times greater than uncatalysed reactions.

All cells do not have the same set of enzymes. Different cells contain different enzymes according to their functions. Many enzymes are dissolved in the cytoplasm, some are bound to organelle membranes, and some are integral parts of ribosomes.

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Active Site

Enzymes are three-dimensional globular proteins. There is a small cleft or depression on the surface called the **active site**, consisting of only a few amino acids. It is the location at which catalysis occurs. The shape of the active site is very specific, so only a certain substrate molecule can fit into it.

The active site has two regions: the **binding site**, where the substrate fits by weak chemical forces such as hydrogen bonds, and the **catalytic site**, which catalyses the reaction and transforms substrate into products.

> **Important**
>
> Some enzymes are produced in inactive form to prevent damage to the cell. For example, pepsin is produced as **pepsinogen** and activated only at its target site.

Important Conceptual Short Questions — Topic 5.1

S1. Why are enzymes called biocatalysts?

They speed up specific chemical reactions by lowering activation energy without being consumed or altered in the process.

S2. How does the active site determine enzyme specificity?

The active site has a specific three-dimensional shape and charge, so only a particular substrate can fit into it and undergo catalysis.

S3. Why is pepsin secreted as pepsinogen?

Pepsin is a protein-digesting enzyme. If active inside the cell, it could destroy cellular proteins, so it is secreted in inactive form and activated at its target site.

S4. Differentiate between the binding site and catalytic site of an enzyme.

The binding site holds the substrate by weak forces, while the catalytic site carries out the chemical reaction that converts substrate into products.

TOPIC 5.2 — COFACTORS AND COENZYMES

Textbook topic 5.2 · printed pages 128–129

Q.2 Describe cofactors and coenzymes and their role in enzyme action.

Board long question · 8 marks

Many enzymes use additional non-protein components called **cofactors** to aid catalysis. The protein part of an enzyme is called the **apoenzyme**, and the complete enzyme including its cofactor is called the **holoenzyme**.

Type	Nature	Role / example
Metal ions	Ca^{2+}, Mg^{2+}, Mn^{2+}, Cu^{2+}, Zn^{2+}	Change non-functional active sites into functional sites and help substrate binding
Prosthetic groups	Cofactors covalently bonded to enzyme; may be organic, e.g., hematin	Permanently associated with enzyme
Coenzymes	Loosely attached non-protein organic molecules	Transport electrons/hydrogen atoms between enzymes; many vitamins function as coenzymes

The most important coenzyme is **NAD^+** (nicotinamide adenine dinucleotide). When NAD^+ accepts a hydrogen atom from an enzyme, it is reduced to **NADH**, carrying energy-rich electrons. Vitamins such as niacin and riboflavin function as coenzymes or parts of coenzymes.

Important Conceptual Short Questions — Topic 5.2

S1. Why do some enzymes require cofactors?

Cofactors help convert non-functional active sites into functional sites and allow the enzyme to combine with its substrate.

S2. Differentiate between apoenzyme and holoenzyme.

The apoenzyme is the protein part of the enzyme. The holoenzyme is the complete, active enzyme including its cofactor.

S3. How do coenzymes differ from prosthetic groups?

Coenzymes are loosely attached organic molecules, while prosthetic groups are covalently bonded to the enzyme and remain permanently associated.

S4. Why is NAD^+ important in cellular respiration?

It acts as a hydrogen acceptor, becoming reduced to NADH and carrying energy-rich electrons from food oxidation to other reactions.

S5. How are vitamins related to enzyme function?

Many vitamins, such as niacin and riboflavin, function as coenzymes or parts of coenzymes, helping enzymes catalyse reactions.

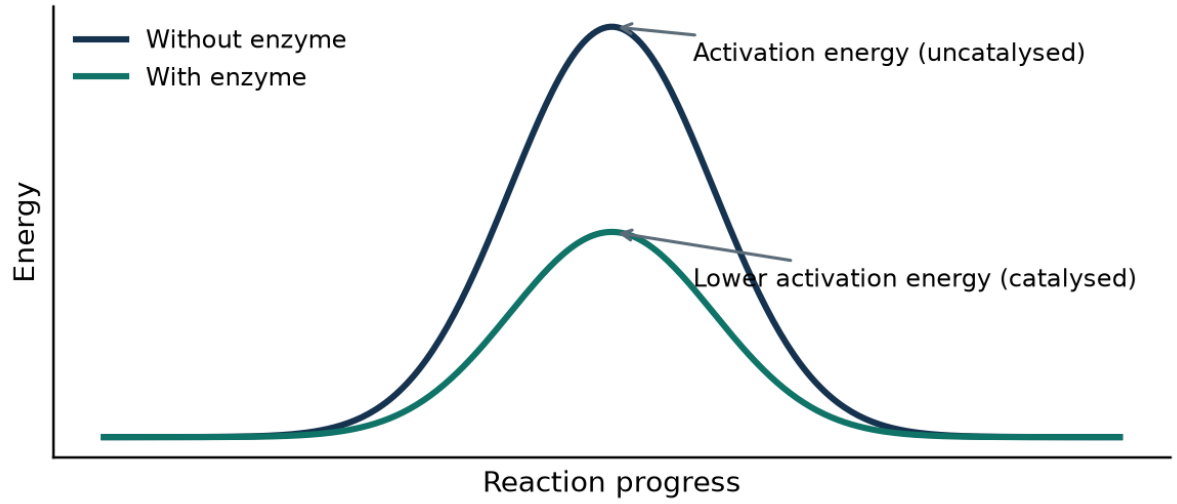
TOPIC 5.3 — MECHANISM OF ENZYME ACTION

Textbook topic 5.3 · printed pages 129–131

Q.3 Explain the mechanism of enzyme action. Compare lock-and-key and induced fit models.

Board long question · 5 + 3 marks

The speed of a chemical reaction depends on the **activation energy** required to initiate it. Enzymes bring reactants together in the correct orientation or stress particular chemical bonds, lowering activation energy and speeding up the reaction.



Enzymes lower the activation energy of a reaction.

The substrate binds to the active site, forming an **enzyme-substrate (ES) complex**. The catalytic site stresses and destabilizes bonds of the substrate, lowering activation energy. The substrate is transformed into products, and the enzyme detaches in an unaltered state.

Enzyme + Substrate → Enzyme-substrate complex → Enzyme + Product

Summary of enzyme action

Model	Proposed by	Main idea
Lock-and-key	Emil Fischer (1894)	Active site is rigid; a specific substrate fits like a key into a lock
Induced fit	Daniel Koshland (1958)	Active site is flexible; substrate binding induces changes that improve catalysis

Important Conceptual Short Questions — Topic 5.3

S1. How do enzymes lower activation energy?

They bring reactants together in the correct orientation and stress particular bonds, reducing the energy needed to start the reaction.

S2. Why does the enzyme remain unaltered after catalysis?

It only facilitates the reaction by forming a temporary enzyme-substrate complex and releases the products without being consumed.

S3. Differentiate between lock-and-key and induced fit models.

Lock-and-key proposes a rigid active site that exactly fits the substrate. Induced fit proposes a flexible active site that changes shape upon substrate binding.

S4. Why is the induced fit model more widely accepted?

It explains how enzymes can adjust their shape to bind substrates more effectively and catalyse reactions, which the rigid lock-and-key model cannot fully explain.

S5. How does feedback inhibition regulate metabolic pathways?

The end product of a pathway inhibits the first enzyme, preventing overproduction and regulating the rate of the pathway.

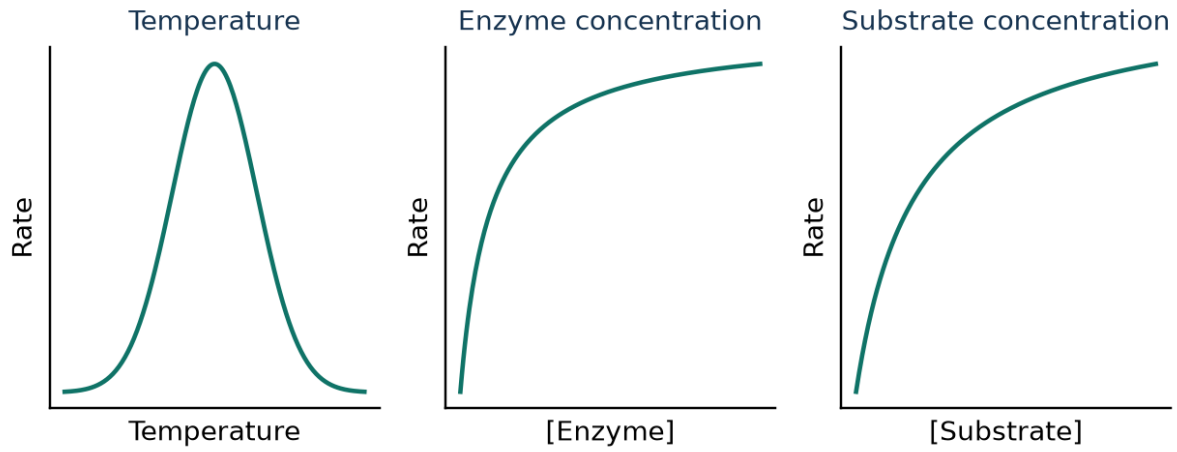
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TOPIC 5.4 — FACTORS AFFECTING THE RATE OF ENZYME ACTION

Textbook topic 5.4 · printed pages 131–133

Q.4 Describe the factors affecting the rate of enzyme action.

Board long question · 8 marks



Effects of temperature, pH and substrate concentration on enzyme activity.

- **Temperature:** Every enzyme has an optimum temperature (37°C for human enzymes). Below optimum, bonds become less flexible and reaction rate is slow. Above optimum, heat disrupts bonds and causes **denaturation**, rapidly decreasing activity.
- **pH:** Each enzyme has an optimum pH. Pepsin works in acidic medium, trypsin in alkaline medium. Changes in pH alter ionization of amino acids at the active site and can denature the enzyme.
- **Enzyme concentration:** If substrate is unlimited, increasing enzyme concentration increases reaction rate because more active sites are available. If substrate is limited, the rate plateaus.
- **Substrate concentration:** Increasing substrate increases rate until all active sites are occupied. Beyond this point, further substrate does not increase the rate.

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Important Conceptual Short Questions — Topic 5.4

S1. Why does enzyme activity decrease above optimum temperature?

High temperature disrupts hydrogen bonds and hydrophobic interactions, causing denaturation and loss of the enzyme's three-dimensional shape.

S2. How does pH affect enzyme activity?

pH changes the ionization of amino acids at the active site and can break bonds holding the enzyme's structure, reducing or blocking activity.

S3. Why does increasing substrate concentration eventually stop increasing reaction rate?

All active sites become occupied, so additional substrate molecules cannot bind and the rate reaches a maximum.

S4. Why do thermophilic bacteria have enzymes that function at high temperatures?

Their proteins have stronger bonding between polypeptide chains, allowing them to remain stable and functional at 70°C or higher.

S5. Differentiate between the effect of enzyme concentration and substrate concentration on reaction rate.

Increasing enzyme concentration increases rate if substrate is unlimited. Increasing substrate concentration increases rate only until all enzyme active sites are saturated.

TOPIC 5.5 — ENZYME INHIBITION

Textbook topic 5.5 · printed pages 133–135

Q.5 Describe enzyme inhibition and its types. How is it significant in medicine and metabolism?

Board long question · 8 marks

A chemical that interferes with and blocks an enzyme's activity is called an **inhibitor**. Inhibitors attach to enzymes but are not transformed into products, blocking active sites temporarily or permanently.

Type	Mechanism	Example
Competitive inhibitor	Resembles substrate and competes for the active site	Malonic acid inhibits succinic dehydrogenase
Non-competitive inhibitor	Binds at a site other than the active site and alters enzyme shape	Succinyl-CoA inhibits succinyl-CoA synthetase
Reversible inhibitor	Forms weak bonds; inhibition can be reversed	Malonate
Irreversible inhibitor	Forms covalent bonds; cannot be easily reversed	Penicillin, cyanide

Significance: Enzyme inhibition regulates metabolic pathways, is the basis of many drugs (antibiotics, anticancer drugs, anticoagulants), helps manage poisoning, and serves as a tool in pharmaceutical research. Aspirin inhibits prostaglandin-producing enzymes, reducing pain and inflammation.

Important Conceptual Short Questions — Topic 5.5

S1. How does a competitive inhibitor block enzyme activity?

It resembles the substrate and competes for the same active site, preventing substrate binding.

S2. Why does a non-competitive inhibitor not compete with the substrate?

It binds at a different site and changes the enzyme's shape, so the active site no longer fits the substrate.

S3. Differentiate between reversible and irreversible inhibitors.

Reversible inhibitors form weak bonds and can be removed, restoring enzyme activity. Irreversible inhibitors form covalent bonds and permanently disable the enzyme.

S4. How does aspirin act as an enzyme inhibitor?

It inhibits enzymes that produce prostaglandins, thereby suppressing pain and inflammation.

S5. Why is cyanide poisonous?

It irreversibly inhibits cytochrome oxidase by binding to copper and iron in its active site, blocking cellular respiration.

S6. How does enzyme inhibition help regulate metabolic pathways?

End products can inhibit enzymes of earlier steps, controlling the rate of the pathway and preventing overproduction.

TOPIC 5.6 — CLASSIFICATION OF ENZYMES

Textbook topic 5.6 · printed pages 135–137

Q.6 Classify enzymes on the basis of reactions and substrates.

Board long question · 8 marks

Classification on the basis of reactions

Class	Reaction catalysed	Example
Oxidoreductases	Oxidation/reduction; add or remove H^+ or electrons	Cytochrome oxidase
Transferases	Transfer functional groups such as methyl, acyl, amino or phosphate	Hexokinase
Hydrolases	Hydrolysis; break substrates by adding water	Lipase, amylase, peptidase
Lyases	Non-hydrolytic addition or removal of groups such as CO_2	Pyruvate decarboxylase
Isomerases	Intramolecular rearrangement; convert one isomer to another	Hexose isomerase
Ligases	Join two molecules using ATP energy	Polymerases

Classification on the basis of substrates

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Group	Substrate	Examples
Proteases	Proteins	Pepsin, trypsin, aminopeptidases, erepsin
Lipases	Lipids	Pancreatic lipase
Carbohydrases	Carbohydrates	Amylase, cellulase, maltase, sucrase, lactase
Nucleases	Nucleic acids	RNAase, DNAase, ATPase

Important Conceptual Short Questions — Topic 5.6

S1. Why are oxidoreductases important in cellular respiration?

They catalyse oxidation-reduction reactions, transferring electrons and hydrogen ions during energy release from food molecules.

S2. How do hydrolases differ from lyases?

Hydrolases break bonds by adding water, while lyases add or remove groups without hydrolysis.

S3. Why do ligases require ATP?

They join two molecules by forming new bonds, which requires energy supplied by ATP.

S4. Differentiate between transferases and isomerases.

Transferases move functional groups from one molecule to another. Isomerases rearrange atoms within a molecule to convert one isomer into another.

S5. How does substrate-based classification help in understanding digestion?

It groups enzymes by the food molecules they break down: proteases for proteins, lipases for lipids, carbohydrases for carbohydrates and nucleases for nucleic acids.

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